

Software Design in Group

EER4189

PROJECT PROPOSAL REPORT

Service Station Management System for Omega Engineers Service Station

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Submission Date	: 26.08.2026

Document Control

Version	Date	Description	Prepared By	Approval
1.0	20 August 2026	View project Proposal	Team CodeLab	Client
1.2	21 August 2026	Corrected project proposal	Team CodeLab	Supervisor

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Introduction and Background

1.1 Introduction

Information technology has become an essential component in modern business operations. Organizations increasingly adopt digital systems to improve efficiency, reduce operational errors, and enhance customer satisfaction. Omega Engineers Service Station provides vehicle repair, maintenance, mechanical engineering services, spare parts handling, and customer billing services. Currently, many of these business processes are carried out manually, making daily operations time-consuming and difficult to manage effectively.

This project proposes the development of a **Service Station Management System** to digitize and streamline the service station's operations through a centralized, web-based platform.

1.2 Background

Omega Engineers Service Station is in Niyamgamdora, Kothmale and serves customers requiring vehicle maintenance and repair services. The service station manages customer information, vehicle records, repair jobs, billing details, and spare-part inventories. At present, records are maintained using paper-based methods, which creates operational difficulties and increases the risk of data loss and human errors.

The proposed system aims to modernize the service station by introducing a digital solution that improves workflow management, data accuracy, service quality, and decision-making.

2. Problem Statement and Project Objectives

2.1 Problem Statement

According to the client, Omega Engineers Service Station relies heavily on manual paper-based processes to manage job cards, appointments, inventory, and customer communication. This creates several operational issues that reduce efficiency and affect service delivery.

- Misplaced or incomplete job cards
- Slow customer service processes
- Difficulty tracking spare parts inventory
- Absence of a centralized information system
- Lack of proper reporting facilities for management
- Delays in accessing service records and business information

2.2 Main Objective

To develop a centralized **Service Station Management System** that automates and improves the operational activities of Omega Engineers Service Station.

2.3 Specific Objectives

- Digitize the job card management process
- Improve appointment scheduling and management
- Maintain accurate inventory records
- Automate billing and invoice generation
- Improve mechanic task assignment and tracking
- Provide management reports for decision-making
- Establish a centralized database for operational information

3. Similar Existing Systems

Several service station and workshop management systems are available in the market. These systems demonstrate similar functionality that can guide the design of the proposed solution.

System	Main Features	Relevance to Proposed System
Shopmonkey	Repair order management, customer handling, invoicing, and reporting	Provides a useful reference for workflow and billing features
Tekmetric	Vehicle service tracking, customer communication, workflow management, and inventory control	Shows the importance of job tracking and inventory visibility
AutoFluent	Scheduling, inventory management, customer management, accounting support, and reporting	Highlights integrated service station management capabilities

Although existing systems provide useful functionality, they are generic commercial solutions. The proposed Service Station Management System will be customized according to the operational needs of Omega Engineers Service Station, making it more suitable for the client's environment.

4. Proposed Solution

Module	Description
Digital Job Card Management	Create electronic job cards, store repair details, monitor repair progress, and track completed or pending work.
Appointment Scheduling	Manage customer appointments, reduce scheduling conflicts, maintain service schedules, and improve customer convenience.
Inventory Tracking	Manage spare-part information, monitor inventory availability, update stock records, and identify low-stock items.
Mechanic Task Assignment	Assign repair jobs to mechanics, track assigned work, and monitor completion status.
Billing and Invoice Generation	Generate invoices automatically, maintain customer payment records, and reduce billing errors.
Reporting Dashboard	Generate daily service reports, inventory reports, financial summaries, and operational performance statistics.

4.1 High-Level Use Case Diagram

The following high-level use case view summarizes how the main users interact with the proposed Service Station Management System. A more detailed usecase description is provided in the Preliminary SRS appendix.

Actor	Main Use Cases
Administrator / Owner	Login; manage internal users; manage customer and vehicle records; record appointments; create job cards; assign mechanics; manage inventory; generate invoices; record payments; view reports.
Receptionist / Service Advisor	Login; manage customer records; manage vehicle records; record appointments; create job cards; view job status.
Mechanic	Login; view assigned jobs; update repair status.
Inventory / Billing Staff	Login; manage inventory; generate invoices; record payments.

High-Level Use Case Diagram Summary for the Proposed System

5. Technology Plan

Category	Technology / Tool
Backend Technologies	PHP, Laravel, Node.js
Frontend Technologies	HTML5, CSS3, JavaScript, React, Nuxt.js, Flutter
Database	MySQL, PostgreSQL
UI/UX Design	Figma
Development Tools	Visual Studio Code, GitHub, Git, Redis, Docker, Nginx
Browser Support	Google Chrome, Mozilla Firefox, Microsoft Edge
Generative AI tools planned to be used	Microsoft Copilot, Gemini, Claude AI

The project team will follow an **Agile development approach** to support iterative implementation, continuous testing, and regular feedback from the supervisor and client.

6. Project Timeline and Conclusion

6.1 Project Timeline

No.	Date	Activity / Milestone
1.	25 June 2026	Choose client requirement
2.	30 June 2026	Upload Proposed Project Slides
3.	26 August 2026	Project Proposal Report Submission
4.	10 October 2026	Progress Review Presentation
5.	10 October 2026	Project Progress Report Submission
6.	01 December 2026	Final Project Report Submission
7.	01 December 2026	Final Project Demonstration

6.2 Conclusion

The Service Station Management System is proposed to improve the operational efficiency of Omega Engineers Service Station. The system will replace manual paper-based processes with a centralized digital solution that supports job card management, appointment scheduling, inventory tracking, mechanic task management, billing, and reporting. The implementation of the system is expected to enhance productivity, improve customer service, and provide accurate information for management decision-making.

Appendix A: Preliminary Software Requirements Specification (SRS)

A.1 Introduction

Purpose: The purpose of this Software Requirements Specification (SRS) is to clearly define the functional and non-functional requirements of the proposed **Service Station Management System** for Omega Engineers Service Station. This document describes what the system will do, how different users will interact with it, and the quality requirements that must be satisfied during development. It provides a common reference for the project team, supervisor, and client to ensure that the system is developed according to the agreed requirements.

Project Overview: The proposed system is a centralized, web-based application designed to support the daily operations of Omega Engineers Service Station. It will help authorized users manage customer records, vehicle records, service appointments, digital job cards, mechanic assignments, repair status updates, spare-part inventory, invoice generation, payment recording, and management reports. The main users of the system are the **Administrator / Owner, Receptionist / Service Advisor, Mechanic, and Inventory / Billing Staff**. By replacing manual paper-based processes with a structured digital solution, the system is expected to improve data accuracy, reduce delays, support better communication between staff members, and provide reliable information for decision-making.

A.2 Functional Requirements

User Story ID	Related Persona	User Story	Acceptance Criteria	Priority
US01	All authorized users	As an authorized system user, I want to log in securely so that I can access only the functions allowed for my role.	Valid credentials allow access; invalid credentials are rejected; the system displays a role-based dashboard.	Must
US02	Administrator / Owner	As an Administrator / Owner, I want to manage internal user accounts so that staff access can be controlled securely.	The Administrator / Owner can create, update, deactivate, and assign roles to staff accounts.	Must
US03	Receptionist / Service Advisor	As a Receptionist / Service Advisor, I want to manage customer records so that customer information is stored accurately in one centralized system.	The actor can add, search, update, and save customer records without creating duplicate entries.	Must

US04	Receptionist / Service Advisor	As a Receptionist / Service Advisor, I want to manage vehicle records so that each customer vehicle can be linked to service and repair history.	The actor can add, update, and search vehicle details linked to the correct customer.	Must
US05	Receptionist / Service Advisor	As a Receptionist / Service Advisor, I want to record appointments so that customer service visits can be scheduled properly.	The actor can enter customer, vehicle, service, date, and time details; duplicate bookings are prevented.	Must
US06	Receptionist / Service Advisor	As a Receptionist / Service Advisor, I want to create job cards so that vehicle repair details can be recorded and tracked digitally.	The actor can create a job card linked to a customer and vehicle; the system generates a unique job card ID.	Must
US07	Administrator / Owner	As an Administrator / Owner, I want to assign mechanics to job cards so that repair work is clearly allocated to responsible staff members.	The Administrator / Owner can select an available mechanic; the assignment is saved and visible to the mechanic.	Must
US08	Receptionist / Service Advisor	As a Receptionist / Service Advisor, I want to view job status so that I can respond accurately to customer inquiries.	The actor can view pending, in-progress, and completed job statuses linked to customer job cards.	Must
US09	Mechanic	As a Mechanic, I want to view assigned jobs so that I can understand my daily repair responsibilities.	The mechanic can view only assigned job cards with vehicle details, repair descriptions, and current status.	Must
US10	Mechanic	As a Mechanic, I want to update repair status so that service progress can be monitored by front-office staff and management.	The mechanic can update status as pending, in progress, or completed; updates are saved and visible to authorized staff.	Must
US11	Inventory / Billing Staff	As Inventory / Billing Staff, I want to manage spare-part inventory so that stock levels remain accurate.	The actor can add, update, and monitor spare-part records; low-stock items are clearly highlighted.	Must
US12	Inventory / Billing Staff	As Inventory / Billing Staff, I want to generate invoices so that billing can be completed accurately after services are performed.	The system calculates service charges and spare-part costs; invoices contain customer, vehicle,	Must

			job card, and total payment details.	
US13	Inventory / Billing Staff	As Inventory / Billing Staff, I want to record payments so that invoice payment status can be maintained accurately.	The actor can enter payment details; the system updates invoice payment status and stores the payment record.	Must
US14	Administrator / Owner	As an Administrator / Owner, I want to view reports so that I can make management decisions using accurate service, inventory, and payment information.	The Administrator / Owner can generate reports by selected period and view service, inventory, payment, and performance summaries.	Should

A.3 Non-Functional Requirements

NFR Category	Requirement ID	Non-Functional Requirement
Performance	NFR01	The system shall respond to normal user operations within 3 seconds under standard operating conditions.
	NFR02	The system shall support multiple users accessing customer, job card, inventory, and billing functions without noticeable delay.
Security	NFR03	The system shall authenticate users using valid login credentials before allowing access to system functions.
	NFR04	The system shall implement role-based access control to restrict functions according to user roles such as Administrator / Owner, Receptionist / Service Advisor, Mechanic, and Inventory / Billing Staff.
Reliability	NFR05	The system shall maintain data integrity when storing customer records, job cards, inventory details, invoices, and reports.
	NFR06	The system shall reduce the risk of data loss by storing information in a centralized database rather than relying on paper-based records.
Usability	NFR07	The system shall provide a simple and user-friendly interface that can be understood by Administrator / Owner, Receptionist / Service Advisor, Mechanic, and Inventory / Billing Staff users with minimal training.

	NFR08	The system shall use clear labels, menus, buttons, and forms to help users complete tasks such as booking appointments, creating job cards, and generating invoices.
Compatibility	NFR09	The system shall operate correctly on major modern web browsers including Google Chrome, Mozilla Firefox, and Microsoft Edge.
	NFR10	The system interface shall be responsive enough to support common desktop and laptop screen sizes used by the service station.

A 4. Use Case Diagram Description

Use Case ID	UC01
Use Case Name	Login
Description	Allows authorized users to access the system according to their assigned user role.
Primary Actor	Administrator / Owner, Receptionist / Service Advisor, Mechanic, Inventory / Billing Staff
Preconditions	The user must have a registered account with valid login credentials.
Postconditions	The user is redirected to the role-based dashboard after successful authentication.
Main Flow	1. The user opens the login page. 2. The user enters username and password. 3. The system validates the credentials. 4. The system identifies the user role. 5. The system displays the appropriate dashboard.
Alternative Flow	If the user forgets the password, the system may allow password reset through an administrator-controlled process.
Exception Flow	If invalid credentials are entered, the system displays an error message and denies access.
Traceable User Stories	US01

Use Case ID	UC02
Use Case Name	Manage Internal Users
Description	Allows the Administrator / Owner to create, update, deactivate, and manage staff user accounts and their role-based access permissions.
Primary Actor	Administrator / Owner
Preconditions	The Administrator / Owner must be logged in and have permission to manage system users.
Postconditions	Staff account details and access permissions are saved or updated successfully.
Main Flow	1. The Administrator / Owner opens user management. 2. The Administrator / Owner adds or selects a staff account. 3. The

	Administrator / Owner enters user details and assigns a role. 4. The system validates the details. 5. The system saves the user account.
Alternative Flow	If a staff member already exists, the Administrator / Owner updates the existing account instead of creating a duplicate account.
Exception Flow	If required account details are missing or invalid, the system displays validation messages and prevents saving.
Traceable User Stories	US02

Use Case ID	UC03
Use Case Name	Manage Customer Records
Description	Allows authorized staff to add, search, update, and maintain customer information in the system.
Primary Actor	Receptionist / Service Advisor, Administrator / Owner
Preconditions	The actor must be logged in and have permission to manage customer records.
Postconditions	Customer information is stored or updated successfully in the centralized database.
Main Flow	1. The actor selects customer management. 2. The actor enters or searches customer details. 3. The system validates required information. 4. The actor saves the record. 5. The system confirms successful storage.
Alternative Flow	If the customer already exists, the actor updates the existing record instead of creating a duplicate record.
Exception Flow	If mandatory details are missing, the system displays validation messages and prevents saving.
Traceable User Stories	US03

Use Case ID	UC04
Use Case Name	Manage Vehicle Records
Description	Allows authorized staff to add, update, search, and maintain vehicle details linked to customer records.
Primary Actor	Receptionist / Service Advisor, Administrator / Owner
Preconditions	The actor must be logged in and the related customer record must be available.
Postconditions	Vehicle details are stored or updated and linked to the correct customer record.
Main Flow	1. The actor opens vehicle management. 2. The actor selects the related customer. 3. The actor enters or updates vehicle details. 4. The system validates the details. 5. The system saves the vehicle record.
Alternative Flow	If the vehicle already exists, the actor updates the existing vehicle record instead of creating a duplicate record.
Exception Flow	If mandatory vehicle information is missing, the system displays validation messages and prevents saving.
Traceable User Stories	US04

Use Case ID	UC05
Use Case Name	Record Appointment
Description	Allows the Receptionist / Service Advisor or Administrator to record a customer's vehicle service appointment.
Primary Actor	Receptionist / Service Advisor, Administrator / Owner
Preconditions	The actor must be logged in, and customer details must be available or entered during the process.
Postconditions	The appointment is saved and becomes visible in the service schedule.
Main Flow	1. The actor selects record appointment. 2. The actor enters customer, vehicle, service, date, and time details. 3. The system checks appointment availability. 4. The actor confirms the booking. 5. The system saves the appointment.
Alternative Flow	If the requested time slot is unavailable, the actor selects another available date or time.
Exception Flow	If appointment details are incomplete or invalid, the system displays an error message and requests correction.
Traceable User Stories	US05

Use Case ID	UC06
Use Case Name	Create Job Card
Description	Allows authorized staff to create a digital job card for a vehicle repair or service request.
Primary Actor	Receptionist / Service Advisor, Administrator / Owner
Preconditions	The actor must be logged in, and customer and vehicle details must be available.
Postconditions	A job card is created with a unique ID and becomes available for assignment and tracking.
Main Flow	1. The actor selects create job card. 2. The actor selects customer and vehicle records. 3. The actor enters repair complaints and service details. 4. The actor assigns a mechanic or forwards the job for assignment. 5. The system saves the job card.
Alternative Flow	If the job card is based on an existing appointment, appointment details are reused to reduce duplicate data entry.
Exception Flow	If required job details are missing, the system prevents job card creation and displays validation messages.
Traceable User Stories	US06, US07

Use Case ID	UC07
Use Case Name	Assign Mechanic
Description	Allows the Administrator / Owner to assign an available mechanic to a selected job card.
Primary Actor	Administrator / Owner
Preconditions	The Administrator / Owner must be logged in, a job card must exist, and at least one mechanic must be available.
Postconditions	The selected mechanic is assigned to the job card and can view the assigned job.
Main Flow	1. The Administrator / Owner opens the job card. 2. The Administrator / Owner selects assign mechanic. 3. The system displays available mechanics. 4. The Administrator / Owner selects a mechanic. 5. The system saves the assignment.
Alternative Flow	If the preferred mechanic is unavailable, the Administrator / Owner selects another available mechanic.
Exception Flow	If no mechanic is available, the system displays an availability message and does not complete the assignment.
Traceable User Stories	US07

Use Case ID	UC08
Use Case Name	Manage Inventory
Description	Allows authorized users to manage spare-part records and stock levels.
Primary Actor	Inventory / Billing Staff, Administrator / Owner
Preconditions	The actor must be logged in and have inventory management permission.
Postconditions	Inventory records are updated and stock levels are accurate.
Main Flow	1. The actor selects inventory management. 2. The actor adds, updates, or removes spare-part details. 3. The system validates stock data. 4. The actor saves the changes. 5. The system updates inventory records.
Alternative Flow	If stock reaches the minimum level, the system highlights the item as low stock.
Exception Flow	If invalid quantity values are entered, the system rejects the update and requests correction.
Traceable User Stories	US11

Use Case ID	UC09
Use Case Name	Generate Invoice
Description	Allows authorized billing staff to generate an invoice for completed service work using job card, service, and spare-part details.
Primary Actor	Inventory / Billing Staff, Administrator / Owner
Preconditions	The actor must be logged in, and a valid job card with service and spare-part details must exist.
Postconditions	The invoice is generated, saved, and made available for payment recording.

Main Flow	1. The actor selects a completed job card. 2. The system retrieves customer, vehicle, service, and spare-part details. 3. The system calculates service charges, spare-part costs, and total amount. 4. The actor reviews the invoice details. 5. The actor confirms and saves the invoice.
Alternative Flow	If a discount is approved, the actor applies the discount before finalizing the invoice.
Exception Flow	If required job card, service, or inventory details are incomplete, the system displays an error message and prevents invoice generation until the details are corrected.
Traceable User Stories	US12

Use Case ID	UC10
Use Case Name	Record Payment
Description	Allows authorized billing staff to record customer payment details against a generated invoice.
Primary Actor	Inventory / Billing Staff, Administrator / Owner
Preconditions	The actor must be logged in, and a generated invoice must be available for the customer job.
Postconditions	The payment record is saved and the invoice payment status is updated.
Main Flow	1. The actor opens the invoice. 2. The actor selects record payment. 3. The actor enters payment amount, payment method, and payment date. 4. The system validates the payment details. 5. The system saves the payment and updates invoice status.
Alternative Flow	If the customer makes a partial payment, the system records the paid amount and keeps the remaining balance as pending.
Exception Flow	If the payment amount or required payment details are invalid, the system displays an error message and prevents saving.
Traceable User Stories	US13

Use Case ID	UC11
Use Case Name	View Reports
Description	Allows the Administrator / Owner to view service, inventory, payment, and business performance reports.
Primary Actor	Administrator / Owner
Preconditions	The Administrator / Owner must be logged in and report data must be available.
Postconditions	The selected report is displayed for management review and decision-making.
Main Flow	1. The Administrator selects reports. 2. The Administrator chooses report type and period. 3. The system retrieves relevant data. 4. The system displays the report summary. 5. The Administrator reviews the report.
Alternative Flow	The Administrator may change filters such as date range or report type to view a different report.

Exception Flow	If no data is available for the selected period, the system displays a no-results message.
Traceable User Stories	US14

Use Case ID	UC12
Use Case Name	View Job Status
Description	Allows the Receptionist / Service Advisor to view the current status of customer job cards and respond to customer inquiries.
Primary Actor	Receptionist / Service Advisor
Preconditions	The actor must be logged in and job card records must be available.
Postconditions	The actor can view the current job status and provide accurate service progress information.
Main Flow	1. The actor opens job status. 2. The actor searches by customer, vehicle, or job card number. 3. The system retrieves the matching job card. 4. The system displays the current job status.
Alternative Flow	If multiple matching records are found, the actor selects the correct job card from the list.
Exception Flow	If no matching job card is found, the system displays a no-results message.
Traceable User Stories	US08

Use Case ID	UC13
Use Case Name	View Assigned Jobs
Description	Allows mechanics to view repair jobs assigned to them.
Primary Actor	Mechanic
Preconditions	The mechanic must be logged in and at least one job must be assigned to the mechanic.
Postconditions	The mechanic can view job details needed to begin or continue repair work.
Main Flow	1. The mechanic logs in. 2. The mechanic selects assigned jobs. 3. The system displays assigned job cards. 4. The mechanic opens a job card to view vehicle and repair details.
Alternative Flow	If no jobs are assigned, the system displays a message indicating that there are no current assignments.
Exception Flow	If the mechanic attempts to access an unassigned job, the system denies access.
Traceable User Stories	US09

Use Case ID	UC14
Use Case Name	Update Repair Status
Description	Allows mechanics to update the progress status of assigned repair jobs.
Primary Actor	Mechanic
Preconditions	The mechanic must be logged in and must have an assigned job card.
Postconditions	The repair status is updated and visible to authorized staff.
Main Flow	1. The mechanic opens assigned jobs. 2. The mechanic selects a job card. 3. The mechanic updates the repair status. 4. The system saves the status update. 5. The updated status becomes visible to staff who monitor job progress.
Alternative Flow	The mechanic may add a short repair note when updating the status.
Exception Flow	If the system fails to save the status update, it displays an error message and keeps the previous status unchanged.
Traceable User Stories	US10

A 5. Assumptions and Constraints

Assumptions

- The client will provide accurate and complete information about the current service station processes.
- Authorized staff members will have basic computer literacy to use the system with minimal training.
- The system will be used mainly by internal staff members: Administrator / Owner, Receptionist / Service Advisor, Mechanic, and Inventory / Billing Staff.
- Customer, vehicle, job card, inventory, invoice, and payment data entered into the system will be accurate.
- The service station will have access to computers or laptops and a suitable internet or local network connection when using the system.
- The client and supervisor will provide feedback during development to confirm requirements and interface design decisions.
- The initial version of the system will focus on the core functions identified in the use case diagram and functional requirements.

Constraints

- The project must be completed within the academic project timeline.
- The system scope is limited to the requirements agreed for Omega Engineers Service Station during the preliminary analysis stage.
- Advanced features such as SMS notifications, email notifications, online customer portal, mobile application, and online payment gateway integration are considered future enhancements.

- The system will not include direct customer login in the initial version unless it is added as a future requirement.
- Development is limited by the selected technologies, available resources, and technical skills of the project team.
- The system must be designed to work on common modern web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge.
- The accuracy of reports depends on the correctness and completeness of data entered by authorized staff.
- System performance may be affected by hardware availability, internet connectivity, local network quality, and hosting environment.

Appendix B: Additional Use Case Diagrams



Figure 1 - Usecase Diagram

Appendix C: Wireframes / UI Mockups

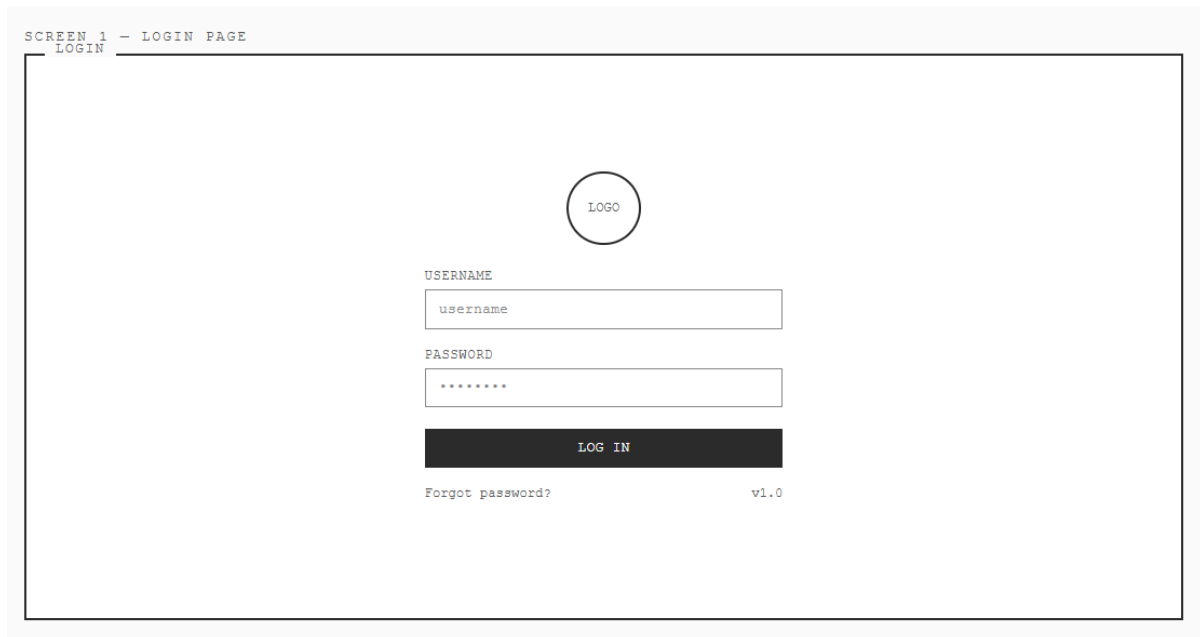


Figure 2 - Login Page



Figure 3 - Dashboard

SCREEN 3 - JOB CARDS PAGE (LIST OF ALL JOB CARDS CREATED FROM THE FORM)
JOB CARDS

+ New Job Card

Filter by Status:
All (12)
Pending (3)
In Progress (5)
Completed (4)

Sort by:

Date Created (newest) ▼

JOB CARD ID	CUSTOMER	VEHICLE NO	MECHANIC	DATE CREATED	STATUS	NEXT SERVICE DATE	
JC-1042	A. Perera	WP CAB-1234	S. Kumara	10 Aug 2026	In Progress	—	View
JC-1041	K. Silva	WP KL-5566	R. Bandara	08 Aug 2026	Completed	08 Feb 2027	View
JC-1040	N. Fernando	WP CAJ-0912	S. Kumara	05 Aug 2026	Pending	—	View
JC-1039	M. Jayawardena	WP GG-3345	R. Bandara	02 Aug 2026	Completed	02 Feb 2027	View

Note: every job card submitted on the Job Card Form (Screen 5) appears here as a new row. "Next Service Date" only fills in once a job is marked Completed.

Figure 4 - Job Cards

SCREEN 5 - JOB CARD FORM
NEW JOB CARD

CUSTOMER & VEHICLE DETAILS

CUSTOMER NAME

e.g. A. Perera

CONTACT NUMBER

07X XXX XXXX

VEHICLE NUMBER

e.g. WP CAB-1234

VEHICLE MODEL

e.g. Toyota Axio

JOB DETAILS

JOB DESCRIPTION

ASSIGN MECHANIC

Select mechanic ▼

STATUS

Completed ▼

SERVICE COMPLETION (FILLED IN WHEN STATUS = COMPLETED)

COMPLETION DATE

08 Aug 2026

NEXT SERVICE DATE

08 Feb 2027

NEXT SERVICE MILEAGE (OPTIONAL)

e.g. 45,000 km

When Status is set to Completed and Next Service Date is entered, the "Save Job Card" button changes to generate the customer's bill - which includes a printable service reminder sticker attached to it (see Screen 6 - Invoice / Bill).

Cancel

Save & Generate Bill →

Figure 5 - New Job Card Form

SCREEN 4 – APPOINTMENTS PAGE (CUSTOMER BOOKINGS, BEFORE A JOB CARD IS OPENED)

Search by customer / vehicle no...

+ New Appointment

Today

This Week

All Upcoming

TIME	CUSTOMER	VEHICLE NO	SERVICE TYPE	STATUS
09:00 AM	A. Perera	WP CAB-1234	Oil Change	Confirmed
11:30 AM	K. Silva	WP KL-5566	Brake Check	Confirmed
02:00 PM	N. Fernando	WP CAJ-0912	Full Service	Pending
04:30 PM	M. Jayawardena	WP GG-3345	Tyre Replacement	Confirmed

Note: clicking "+ New Appointment" opens the popup form on Screen 4b. Once the customer arrives, staff convert the appointment into a Job Card (Screen 5) with one click, carrying over the customer and vehicle details.

Figure 6 - Appointments

SCREEN 4B – NEW APPOINTMENT (POPUP FORM, OPENS ON TOP OF SCREEN 4)

NEW APPOINTMENT

New Appointment

CUSTOMER NAME

e.g. A. Perera

CONTACT NUMBER

07X XXX XXXX

VEHICLE NUMBER

e.g. WP CAB-1234

SERVICE TYPE

Select service ▾

DATE

DD / MM / YYYY

TIME SLOT

Select slot ▾

NOTES (OPTIONAL)

Cancel

Confirm Appointment

System checks the selected Date + Time Slot against existing bookings and blocks it if already taken.

Figure 7 - New Appointment Form

PRINT PREVIEW

Print Preview - JC-1041

Download PDF

Print Bill + Sticker

OMEGA ENGINEERS SERVICE STATION
Solotrade Company · 123 Galle Road, Colombo
Tel: 011 XXX XXXX

INVOICE
No: INV-1041
Date: 08 Aug 2026

BILLED TO
K. Silva · 07X XXX XXXX

VEHICLE
WP KL-5566 · Honda Vezel

JOB CARD NO.
JC-1041

MECHANIC
R. Bandara

ITEM / SERVICE	QTY	UNIT PRICE	AMOUNT
Engine Oil 4L	1	Rs. 3,200	Rs. 3,200
Brake Pad Set	1	Rs. 5,500	Rs. 5,500
Labour Charge	-	-	Rs. 2,000
			Subtotal Rs. 10,700
			Tax Rs. 0
			Total Rs. 10,700

----- %< CUT ALONG LINE - DETACH STICKER BELOW -----

OMEGA ENGINEERS

NEXT SERVICE DUE

Vehicle WP KL-5566

Date 08 Feb 2027

Mileage 45,000 km

Keep this sticker on windshield

Figure 2 - Invoice

Search item...			+ Add Item	
ITEM NAME	QUANTITY	UNIT PRICE	STATUS	
Engine Oil 4L	32	Rs. 3,200	In Stock	
Brake Pad Set	4	Rs. 5,500	Low Stock	
Air Filter	18	Rs. 1,100	In Stock	
Spark Plug	2	Rs. 650	Low Stock	
Coolant 1L	25	Rs. 950	In Stock	

Figure 3 - Inventory

SCREEN 8 - ADD INVENTORY ITEM (OPENS WHEN "+ ADD ITEM" IS CLICKED ON SCREEN 7)
NEW INVENTORY ITEM

ITEM DETAILS

ITEM NAME

e.g. Brake Pad Set

ITEM CODE / SKU

e.g. INV-0234

CATEGORY

Select category ▾

SUPPLIER

Select supplier ▾

STOCK & PRICING

QUANTITY IN STOCK

e.g. 20

UNIT

pcs / litre / box ▾

UNIT PRICE (RS.)

e.g. 5,500

REORDER LEVEL (LOW STOCK ALERT)

e.g. 5

ADDITIONAL INFO

DESCRIPTION / NOTES

ITEM PHOTO (OPTIONAL)

+ Upload image

Cancel

Save Item

Note: once saved, this item appears as a new row in the Inventory table (Screen 6). If Quantity falls at or below Reorder Level, it's auto-tagged "Low Stock" and shown as an alert on the Dashboard.

Figure 10 - Add New Item Form

SCREEN 9 - REPORTS PAGE
REPORTS

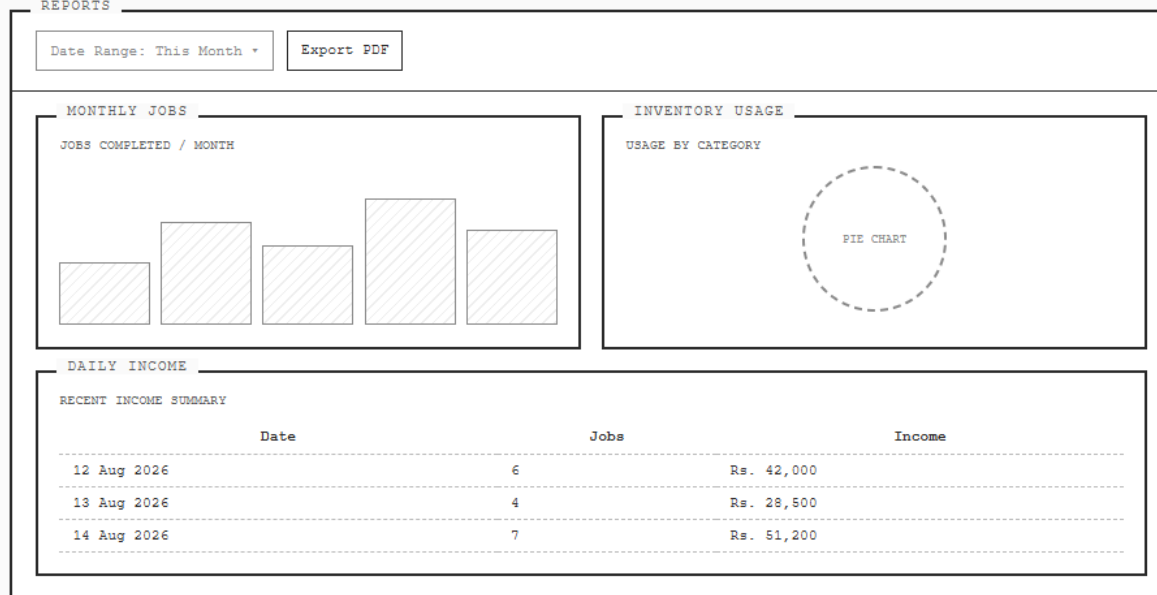


Figure 11 - Report

Appendix D: References and Supporting Materials

Reference / Material	Purpose / Use in Project
Subject Material Book	The subject material book for EER4189 Software Design in Group was used to understand project proposal structure, software requirements specification format, use case modelling, user stories, assumptions and constraints, and documentation guidelines.
LMS Materials and Lecturer Guidance	LMS resources, lecturer instructions, and project guidance were used to understand the required proposal format, appendix requirements, technology planning section, preliminary SRS contents, and submission expectations.
Client Requirement Discussion	Information gathered from Omega Engineers Service Station was used to identify current manual processes, operational problems, required system functions, user roles, and expected improvements.
Similar Existing Systems	Existing service station and workshop management systems such as Shopmonkey, Tekmetric, and AutoFluent were reviewed to understand common features including job card management, appointment scheduling, inventory control, invoicing, and reporting.
Preliminary Wireframes / UI Ideas	Initial interface concepts were used to visualize important screens such as login, dashboard, customer management, job card management, inventory, invoice, and reporting interfaces.

Appendix E: AI Usage Declaration

Generative AI tools were used as supporting tools during the preparation of this project proposal. The project team used AI assistance for brainstorming ideas, improving wording, refining requirement descriptions, checking document structure, and organizing project documentation. AI tools were not used as a replacement for the team's own analysis, decision-making, client requirement gathering, or final academic responsibility.

AI Tool / Platform	Purpose of Use
Microsoft Copilot	Used to support document drafting, wording improvement, structure checking, and requirement refinement.
Gemini	Used to support idea generation, comparison of possible project content, and clarification of technical explanations.
Claude AI	Used to support summarizing, rewriting, and improving clarity of selected proposal content.

All AI-generated suggestions were reviewed, edited, and validated by the project team before being included in the report. The final content, system scope, requirements, diagrams, and conclusions remain the responsibility of the project team.